Yu-Han Huang

estherhan5853@gmail.com | estherhan5853 | Google Scholar | estherhan1.github.io/web | Taipei & Hsinchu, Taiwan

EDUCATION

B.S. in Electrical Engineering, National Taiwan University (NTU), Taipei, Taiwan Sep 2023–June 2027 (expected)

- Selected Coursework: Machine Learning, Deep Learning for Computer Vision, Deep Reinforcement Learning
- Research Interests: Speech and Audio Processing, GUI Agents, Vision-Language-Action Models

RESEARCH EXPERIENCE

Speech Processing and Machine Learning Lab, NTU July 2025–Present

Undergraduate Researcher | Supervisor: Prof. Hung-Yi Lee

- Explored training-free approaches to improve large audio-language model understanding and response reliability
- Investigated safety and alignment methods for full-duplex spoken language models

Natural Language Processing Lab, UCLA July 2026–Present

Visiting Researcher | Supervisor: Prof. Kai-Wei Chang

- Investigated GUI agents for multi-step web navigation and computer-use tasks
- Developed realistic tasks that required GUI agents to combine deep research with multi-step browser interactions

Vision and Learning Lab, NTU Sep 2025–June 2026

Undergraduate Researcher | Supervisor: Prof. Yu-Chiang Frank Wang

- Studied vision-language-action models for long-horizon embodied task planning

INTERNSHIP EXPERIENCE

Synology Inc., New Taipei City, Taiwan Nov 2026–Present

Video Deep Learning Product Developer Intern

Open Cloud Infrastructure Software Center, ASUSTeK Computer Inc., Taipei, Taiwan May 2026–Nov 2026

AI Research Intern

- Conducted research on capability enhancement, evaluation, safety, and alignment for large audio-language models, full-duplex speech models, and LLM-based agents

Institute of Information Science, Academia Sinica, Taipei, Taiwan July 2024–Aug 2024

Student Researcher | Supervisor: Prof. Yennun Huang

- Developed automated pipelines for generating both attack and benign audit logs for intrusion detection research

PUBLICATIONS

** Equal contribution*

[1] Yu-Han Huang, Chih-Kai Yang*, Ke-Han Lu*, Cheng An Yu*, and Hung-yi Lee, "Enhancing How Large Audio-Language Models Hear: Identifying and Amplifying Acoustic Neurons in the Audio Encoder." **[arXiv:2607.11801]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

[2] Kai-Po Chang, Chang-Hsun Wu, Yen-Siang Wu, **Yu-Han Huang**, Kai-Chen Yang, and Yu-Chiang Frank Wang, "Self-Correcting Imagination Planning for VLM Embodied Agents," submitted to the *International Conference on Learning Representations (ICLR 2027)* (under review).

[3] Lok-Lam Ieong*, Chia-Chien Chen*, Chih-Kai Yang*, **Yu-Han Huang**, An-Yu Cheng, and Hung-yi Lee, "Nudging Hidden States: Training-Free Model Steering for Chain-of-Thought Reasoning in Large Audio-Language Models," in *Proc. Interspeech 2026*, 2026, pp. 3797–3802, doi: 10.21437/Interspeech.2026-554.

[4] Kai-Wei Chang, Yi-Cheng Lin, Huang-Cheng Chou, Wenze Ren, **Yu-Han Huang**, Yun-Shao Tsai, Chien-Cheng Chen, Yu Tsao, Yuan-Fu Liao, Shrikanth Narayanan, James Glass, and Hung-yi Lee, "TaigiSpeech: A Low-Resource Real-World Speech Intent Dataset with Scalable Data Mining In-the-Wild," in *Proc. Interspeech 2026* (Long Paper Track), 2026, pp. 7304–7313, doi: 10.21437/Interspeech.2026-1511.

[5] Ke-Han Lu et al., including **Yu-Han Huang**, "How Auditory Knowledge in LLM Backbones Shapes Audio Language Models: A Holistic Evaluation." **[arXiv:2603.19195]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

[6] Yi-Cheng Lin et al., including **Yu-Han Huang**, "Findings of the First Teaching Monster Challenge: A Benchmark of Pedagogical Content Knowledge in AI Agents." **[arXiv:2608.08852]**, submitted to the *AAAI 2027 Artificial Intelligence for Social Impact Track* (under review).

TEACHING EXPERIENCE

Deep Learning for Human Language Processing (CommE5054), EECS Department, NTU — Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Co-organized the First Teaching Monster Challenge and co-developed its baseline pipeline
- Conducted human evaluation of AI-generated teaching videos and coauthored the paper

Machine Learning (EE5184), EECS Department, NTU — Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Designed and prepared the course assignment on model merging

AWARDS & SCHOLARSHIPS

AI Talent+ Overseas Research Scholarship, NTU — July 2026–Aug 2026

- Natural Language Processing Lab, UCLA (Prof. Kai-Wei Chang)

MakeNTU Hackathon – 1st Place, Enterprise Award, Taipei, Taiwan — May 2025

- Project: **AI Driving Coach**
- Integrated WebSocket streaming and the Google Gemini API for driver analysis, voice feedback, and data visualization
- Built an integrated hardware-software architecture to simulate real-world driving scenarios

COURSE PROJECTS

Drifting Q-Learning with Distributional Behavior Cloning — Spring 2026

Deep Reinforcement Learning

- Co-developed DriftQL, an offline reinforcement learning method that replaces pointwise behavior cloning with a distributional drifting objective
- Conducted experiments against 13 baselines on 10 OGBench tasks, achieving state-of-the-art performance on three tasks with one-step inference

Detection-Guided Image Layer Extraction — Fall 2025

Deep Learning for Computer Vision

- Built a two-stage image-decomposition pipeline using Qwen-Image-Layered for RGBA layer generation
- Refined layer boundaries with DocLayout-YOLO-based detection, reducing cross-layer content leakage

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

General Coordinator, NTUEE Night, NTU — March 2026

- Led the department's largest annual event with 200+ performers and staff, attracting an audience of 800+

Director of Public Relations, NTUEE Student Association — Fall 2025–Spring 2026

- Led the Public Relations Department and managed vendor outreach and partnerships for student association events

Vice Captain, NTUEE Women's Basketball Team — Fall 2024

TECHNICAL SKILLS

Programming: Python, C/C++

Deep Learning: PyTorch

LANGUAGES

Mandarin Chinese, English, Japanese (Basic)